

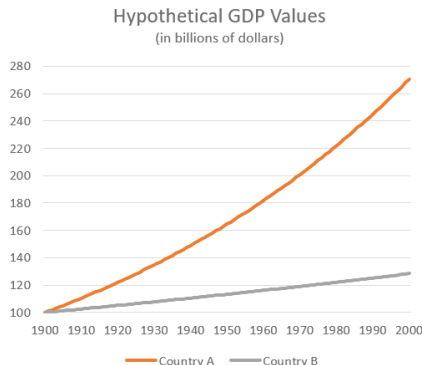
Macroeconomic Theory and Policy: Lecture 3 (Ch.3)

What are You Going to Learn (ch. 3)

- Some facts related to economic growth that later chapters will seek to explain
- The substantial improvement of global welfare brought about by economic growth
- How this growth is a relatively recent phenomenon
- Tools employed to examine economic growth, e.g., growth rate computation
- The rationale behind utilizing a "ratio scale" to facilitate the comprehension of per capita GDP plots

Motivation: Why Is Economic Growth Important?

- A 1 percent change in annual growth appears small, but it may lead to large differences in the levels of output over time
- Consider a 100-year period:
 - ▶ Country A and B begin at the same level of GDP (\$100 billion)
 - ▶ Country A grows at 1 percent each year
 - ▶ Country B grows at 0.25 percent each year



Motivation: Why Is Economic Growth Important?

- Many developed countries of a century ago could be mistaken for a emerging country today
 - ▶ Most of human history has been characterized by little to no economic growth
 - ▶ The last two to three centuries are different than other periods in history in that people lived beyond subsistence levels
- The study of economic growth explains differences between countries and differences over time
 - ▶ For example, why is GDP in the United States higher than in Bangladesh?
 - ▶ Or, why have standards of living increased in the United States over time, despite the severity of the many recessions?

Motivation: Why Is Economic Growth Important?

- Even if a country is experiencing a recession, countries with higher levels of GDP still have higher standards of living
- Economic growth and standards of living are not a short-term result (even though business-cycle fluctuations can affect welfare)

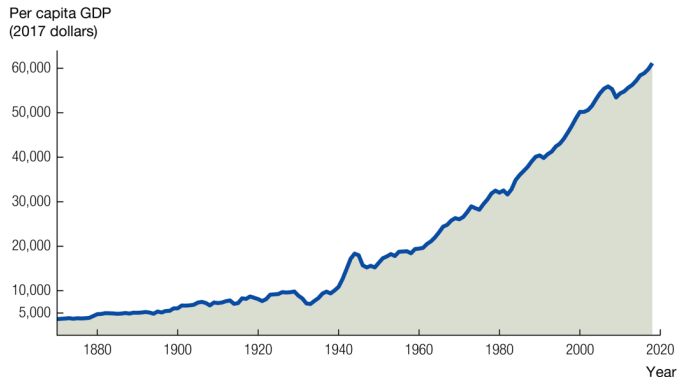
Growth Over the Very Long Run

- Sustained increases in standards of living are a recent phenomenon:
 - ▶ The first agricultural revolution was about 10,000 years ago – humans were hunters and gatherers
 - ★ Wages in ancient Greece and Rome were approximately equal to wages in Britain in the fifteenth century or France in the seventeenth century (prior to modern economic growth)
 - ▶ Yet, only in the last 200–300 years has modern economic growth appeared – urbanization and mechanized
 - ★ 200-300 years is long, but relatively short for human history
- Economic growth emerges in different places at different times for many reasons. Some results of this are:
 - ▶ Standards of living have diverged dramatically
 - ▶ Per capita GDP differs remarkably around the world

Modern Economic Growth

- Looking over the past 150 years: from 1870 to 2018, United States *real* per capita GDP rose by more than 15-fold
 - ▶ On average, around 1870, income per person was around \$3,600 per person (in 2017 dollars)
 - ▶ In 2018, per capita income was about \$61,000.
- Assuming this rate of growth continues, a typical college student today will earn a lifetime income about twice that of his or her parents

Per Capita GDP in the United States, 1870-2020



Source: Data from 1870 to 1928 from Barro- Ursua Macroeconomic Data, 2010. Data from 1929 to 2018 from U.S. Department of Commerce, Bureau of Economic Analysis.
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The Definition of Economic Growth

- Growth of per capita GDP is the rate of change of per capita GDP
 - ▶ The goal is to consider how standards of living have changed over time
 - ▶ We could examine the percentage change in the level of GDP, but GDP per capita gives us a better sense of the average person's income
- A percentage change:

$$\bar{g} = \frac{y_t - y_{t-1}}{y_{t-1}} \text{ so that } y_t = y_{t-1}(1 + \bar{g})$$

where $\bar{g}$ is the associated constant growth rate (i.e., a parameter)

- This tells us that the value of GDP per capita today depends on both the value of GDP per capita in the past and the growth rate of GDP per capita

Example: Population Growth

- Population growth evolves the same way:

$$L_{t+1} = L_t(1 + \bar{n})$$

- Intuitively, tomorrow's population in time period $t + 1$ depends on today's population in period t
- Then, the period after, we have

$$L_{t+2} = L_{t+1}(1 + \bar{n})$$

- Plug in the first equation into the next one

$$L_{t+2} = L_t(1 + \bar{n})(1 + \bar{n}) = L_t(1 + \bar{n})^2$$

Example: Population Growth

- Continue this process for $t + 3$

$$L_{t+3} = L_{t+2}(1 + \bar{n}) = L_{t+1}(1 + \bar{n})(1 + \bar{n}) = L_t(1 + \bar{n})(1 + \bar{n})(1 + \bar{n})$$

That is,

$$L_{t+3} = L_t(1 + \bar{n})^3$$

- In general,

$$L_t = L_0(1 + \bar{n})^t$$

The Constant Growth Rule

- The constant growth rule states that

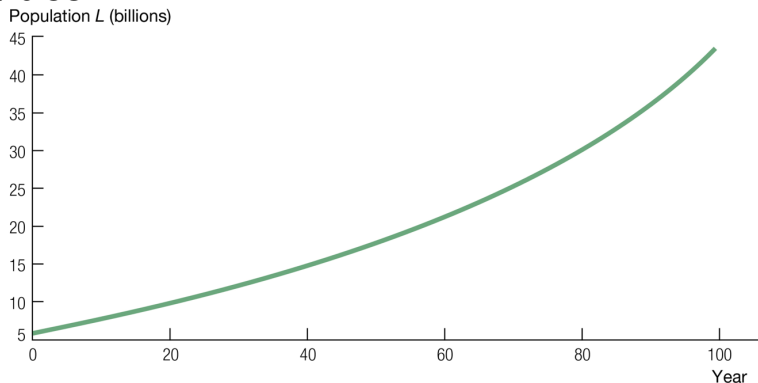
$$y_t = y_0(1 + \bar{g})^t$$

where

- ▶ t is the time period
- ▶ y_t is the value of variable y in time t
- ▶ y_0 is the initial value of variable y in period 0
- ▶ $\bar{g}$ is the constant growth rate

Population over Time

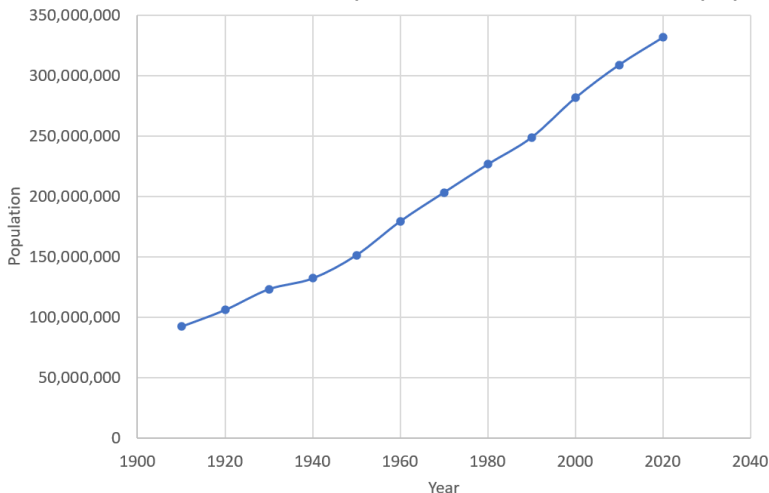
- Using the formula $L_t = L_0(1 + \bar{n})^t$ to compute with $L_0 = 6$ and $\bar{n} = 0.02$ – roughly the birth rate of the US



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Compare to the Actual Population in US

- The shape of the two are similar
- But, the calculation we have miss the important event of WWII and population aging



source: US Census

How to Calculate Constant Growth Rate when it Varies

- For most of the variables, the growth rate varies over time (e.g., growth in GDP)
- Let $g_1 = 5\%$ and $g_2 = 2\%$, to calculate the constant growth rate, we want to find

$$y_0(1 + \bar{g})(1 + \bar{g}) = y_0(1 + 0.05)(1 + 0.02)$$

$$(1 + \bar{g})^2 = (1 + 0.05)(1 + 0.02) \implies \bar{g} = ((1 + 0.05)(1 + 0.02))^{1/2} - 1 = 0.03489$$

It is the geometric average of the growth rate across the two year and is consistent with the way of calculating growth rate

How to Calculate Constant Growth Rate when it Varies

- How's about taking average of the two growth rate using the "usual" method?

$$\bar{g} = \frac{g_1 + g_2}{2} = \frac{0.05 + 0.02}{2} = 0.035$$

The two numbers are close. This is the arithmetic mean.

- The two should be close when the growth rate does not varies too much

Calculating Growth Rates Given Only Two Years of GDP

- What if we are given data only for the start and end of this series?
 - ▶ e.g., United States per capita GDP 1870: \$3,600 and 2018: \$61,000
- Use the constant growth rule: $y_t = y_0(1 + \bar{g})^t$

$$\frac{y_t}{y_0} = (1 + \bar{g})^t \implies \bar{g} = \left(\frac{y_t}{y_0}\right)^{\frac{1}{t}} - 1$$

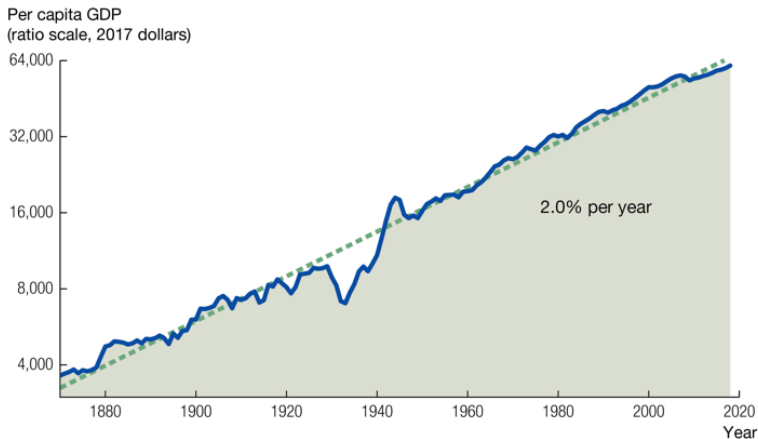
- For example, $\bar{g} = \frac{61000}{3600}^{1/148} - 1 = 0.02$
- For $t = 1$,

$$\bar{g} = \frac{y_1}{y_0} - 1 = \frac{y_1 - y_0}{y_0}$$

The usual percentage change formula

The Ratio (Logarithmic) Scale

- From time to time, you may see a plot where the tick marks are not of equal size



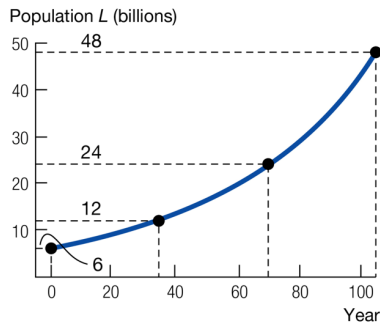
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The Ratio (Logarithmic) Scale

- A plot intervals between tick marks are proportional in terms of ratios rather than absolute differences
 - ▶ When doubling, we want this constant ratio to be "2."
 - ▶ e.g., on the y-axis, instead of "1, 2, 3, 4" we have "1, 2, 4, 8"
- When plotted on a ratio scale, a variable that grows at a constant rate will be a straight line
- It is commonly used to represent data where exponential growth or significant variations need to be visually compressed
 - ▶ Using the population growth figure earlier, we cannot tell right away that the growth rate is constant, increasing, or decreasing

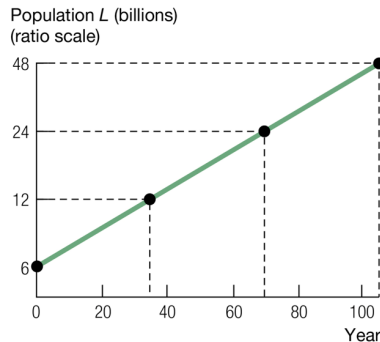
Population over Time, Revisited

- Absolute population numbers are increasing faster (left), but the rate of growth is staying the same in the right graph because of the linear relationship (right)



(a) On a standard scale...

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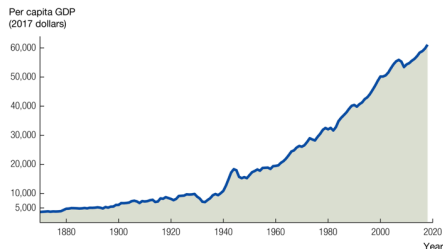
(b) and a ratio scale.

U.S. GDP on a Ratio Scale

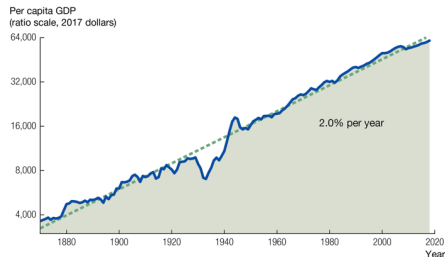
- On a ratio scale:
 - ▶ if growth rates are rising, the slope will be increasing
 - ▶ if the growth rate is constant, it will be a straight line
 - ▶ if the growth rate is falling, the slope will be decreasing
- Per capita GDP in the United States has grown at approximately 2 percent per year over the past 150 years.
 - ▶ This outcome is easy to see using a ratio scale
 - ▶ Approximately linear

Per Capita GDP in the United States, Standard vs. Ratio Scale

- The graph on the left is a standard scale and the one on the right is a ratio scale
- Data series lies close to a line that exhibits a constant slope of 2.0 percent per year
- To a first approximation, per capita GDP in the United States has been growing at a relatively constant annual rate of 2 percent over the past 150 years



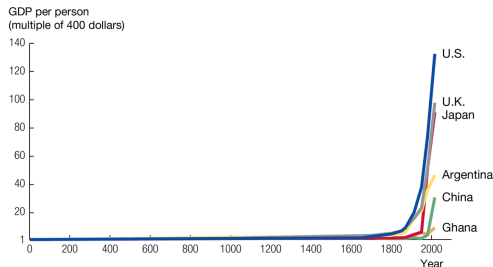
Source: Data from 1870 to 1928 from Barro- Ursua Macroeconomic Data, 2010. Data from 1929 to 2018 from U.S. Department of Commerce, Bureau of Economic Analysis.
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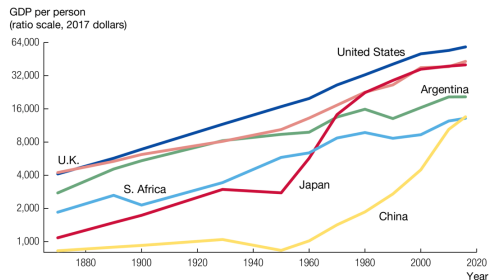
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The Ratio Scale in Action: Per Capita GDP since 1870

- The picture contains more information, but we have to read it carefully
- The per capita income line is flag for the UK relative to the US until the 50s
- From 1880 to 1920, Argentina grew rapidly; Since 1980, growth has slowed considerably



Source: The Maddison-Project, www.ggdc.net/maddison/.
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Source: The Maddison-Project, www.ggdc.net/maddison/. Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.
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Next Lecture

- Next lecture will finish Ch.3 and start covering Ch.4